

# Stationary source emissions — Determination of low range mass concentration of dust —

## Part 2: Automated measuring systems

The European Standard EN 13284-2:2004 has the status of a  
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# National foreword

This British Standard is the official English language version of EN 13284-2:2004.

The UK participation in its preparation was entrusted by Technical Committee EH/2, Air quality, to Subcommittee EH/2/1, Stationary source emissions, which has the responsibility to:

- aid enquirers to understand the text;
- present to the responsible international/European committee any enquiries on the interpretation, or proposals for change, and keep the UK interests informed;
- monitor related international and European developments and promulgate them in the UK.

A list of organizations represented on this subcommittee can be obtained on request to its secretary.

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### Summary of pages

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English version

## Stationary source emissions - Determination of low range mass concentration of dust - Part 2: Automated measuring systems

Emissions de sources fixes - Détermination de la faible concentration en masse de poussières - Partie 2: Systèmes automatiques de mesure

Emissionen aus stationären Quellen - Ermittlung der Staubmassenkonzentration bei geringen Staubkonzentrationen - Teil 2: Automatische Messeinrichtungen

This European Standard was approved by CEN on 22 July 2004.

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## Foreword

This document (EN 13284-2:2004) has been prepared by Technical Committee CEN/TC 264 "Air quality", the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by March 2005, and conflicting national standards shall be withdrawn at the latest by March 2005.

This document has been prepared under a mandate given to CEN by the European Commission and the European Free Trade Association, and supports essential requirements of EU Directive(s).

For relationship with EU Directive(s), see informative bibliography, which is an integral part of this document.

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## Introduction

This document describes the quality assurance procedures related to Automated Measuring Systems (AMS) for the determination of dust in flue gas, in order to meet the uncertainty requirements on measured values given by regulations, e.g. EC Directives ([1], [2]), national or other legislation.

This document is only applicable in conjunction with the general document on quality assurance of AMS described in EN 14181, and provides indications which are specific to dust measurements.

The calibration and validation of dust AMS are performed by parallel measurements with the reference manual gravimetric method described in EN 13284-1.

## 1 Scope

This document specifies specific requirements on automated measuring systems (AMS) for dust monitoring. It is derived from EN 14181 which is the general document on the quality assurance of AMS. It is only applicable in conjunction with EN 14181.

This document specifies test criteria for the QAL1 and specific requirements for QAL2, QAL3 and AST for dust AMS used for proving that the dust emissions from a source are compliant with emission limits below 50 mg/m<sup>3</sup> (standard conditions) in ducted gaseous streams. This document is applicable by direct correlation with the standard reference method (SRM) described in EN 13284-1.

This document is primarily developed for emissions from waste incinerators. From a technical point of view, it may be applied to other processes, for which measurement at an emission limit is required with defined uncertainty.

## 2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

EN 13284-1:2001, *Stationary source emissions - Determination of low range mass concentration of dust - Part 1: Manual gravimetric method*.

EN 14181:2004, *Stationary source emissions – Quality assurance of automated measuring systems*.

EN ISO 14956, *Air quality - Evaluation of the suitability of a measurement procedure by comparison with a required measurement uncertainty (ISO 14956:2002)*.

EN ISO/IEC 17025, *General requirements for the competence of testing and calibration laboratories (ISO/IEC 17025:1999)*.

## 3 Terms and definitions

For the purposes of this document, the terms and definitions given in EN 13284-1:2001 and EN 14181:2004 and the following apply.

### 3.1 dust

particles, of any shape, structure or density, dispersed in the gas phase at the sampling point conditions which may be collected by filtration under specified conditions after representative sampling of the gas to be analysed, and which remain upstream of the filter and on the filter after drying under specified conditions

[EN 13284-1:2001]

### 3.2 automated measuring system (AMS)

measuring system permanently installed on site for continuous monitoring of emissions

NOTE 1 An AMS is a method which is traceable to a reference method.

NOTE 2 Apart from the analyser, an AMS includes facilities for taking samples (e.g. sample probe, sample gas lines, flow meters, regulators, delivery pumps) and for sample conditioning (e.g. dust filter, moisture removal devices, converters, diluters). This definition also includes testing and adjusting devices that are required for regular functional checks.

[EN 14181:2004]

NOTE 3 For the purpose of this document, AMS means automated dust measuring system.

### 3.3

#### cross sensitivity

response of the AMS to determinants other than dust concentration

NOTE Cross sensitivity can be caused for example by particle characteristics (including size, material type, electric charge, shape and moisture content), flow gas conditions (including velocity) and gas composition (including water vapour, water droplets).

### 3.4

#### emission limit value (ELV)

limit value related to the uncertainty requirement

NOTE For the EU directives [1] and [2] it is the daily emission value that relates to the uncertainty requirement.

### 3.5

#### linearity (lack of fit)

systematic deviation, within the range of application, between the accepted value of a reference material applied to the measuring system and the corresponding output signal from the AMS

NOTE Test for linearity is described in Annex B of EN 14181:2004.

### 3.6

#### limit of detection

minimum value of the measurand for which the measuring system is not in the basic state, with a stated probability

NOTE The basic state is usually represented by zero reference material.

### 3.7

#### period of unattended operation

maximum interval of time for which the performance characteristics remain within a predefined range without external servicing, e.g. refill, calibration, adjustment

### 3.8

#### reproducibility under field conditions ( $r$ )

measure of the agreement between two identical measuring systems applied in parallel in field tests at a level confidence of 95 % using the standard deviation of the difference of the paired measurements:

$$r = t_{n-1; 0,95} \times s_D$$

$$s_D = \sqrt{\frac{\sum_{i=1}^n (x_{1,i} - x_{2,i})^2}{2n}}$$

where

$r$  is the reproducibility under field conditions;

$t_{n-1; 0,95}$  is the two-sided Students t-factor at a confidence level of 95 % with a number of degrees of freedom  $n-1$ ;

$s_D$  is the standard deviation of the difference of paired measurements;



- $x_{1,i}$  is the  $i$ th measurement result of the first measuring system;
- $x_{2,i}$  is the  $i$ th measurement result of the second measuring system;
- $n$  is the number of parallel measurements

NOTE The reproducibility under field conditions is calculated from the half-hour averaged output signals during the three-month period.

### 3.9

#### Standard Reference Method (SRM)

method described and standardised to define an air quality characteristic, temporarily installed on site for verification purposes

[EN 14181:2004]

NOTE For the purposes of this document, the SRM is specified in EN 13284-1.

## 4 Principle

This document is derived from the general document on the quality assurance of AMS (EN 14181). It shall be applied in conjunction with that document. This document only deals with the specific aspects related to dust monitoring.

According to EN 14181, three different levels of quality assurance are defined, designated as Quality Assurance Levels (QAL). These levels cover the suitability of an AMS for its measuring task (QAL1), the validation of the AMS following its installation (QAL2), the control of the AMS during its ongoing operation on an industrial plant (QAL3). An annual Surveillance Test (AST) is also defined.

## 5 AMS performance criteria for QAL1

The performance of the AMS shall be assessed in a suitability test referred to as QAL1. Suitability tests for an AMS are usually carried out within the framework of certification or type approval procedures. The suitability test comprises a separate laboratory test and a three-month field test in a typical application.

The requirements for the tests used in QAL1 are given in Table 1. These requirements shall be used until general requirements on AMS are specified in a European Standard on AMS certification.

The manufacturer shall ensure that the delivered AMS has the same characteristics as the tested devices.

Table 1 – QAL1 performance criteria

| <b>LABORATORY TESTS</b>                                                                                                                                                                                                             |                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Zero and span check                                                                                                                                                                                                                 | shall be documented by the manufacturer and assessed during QAL1 to be appropriate for QAL3                                                                                                                 |
| Reference materials                                                                                                                                                                                                                 | shall be documented by the manufacturer and assessed during QAL1 to be appropriate for AST                                                                                                                  |
| Automatic compensation of zero and span drift, if provided                                                                                                                                                                          | shall be documented by the manufacturer and assessed during QAL1;<br>if used as part of QAL3, the AMS shall indicate compensation beyond specified limits                                                   |
| Linearity (lack of fit)                                                                                                                                                                                                             | shall be less than $\pm 4\%$ of ELV and checked by use of reference materials (also to be applied in AST)                                                                                                   |
| Detection limit                                                                                                                                                                                                                     | shall be less than 2 % of ELV                                                                                                                                                                               |
| Influence of physical environment, such as:<br>- temperature<br>- electrical supply<br>- ambient light<br>- optical alignment<br>- vibrations<br>- pressure                                                                         | shall have an individual and total combined influence less than 2 % of ELV within the range of variations defined by the manufacturer;<br>shall be documented by the manufacturer and validated during QAL1 |
| Cross sensitivity (process condition) such as:<br>- ambient light<br>- particle material<br>- particle sizes<br>- particle shapes<br>- velocity<br>- electrical charge<br>- gaseous components including water vapour<br>- droplets | shall be documented by the manufacturer and validated during QAL1                                                                                                                                           |
| <b>FIELD TESTS (3 months)</b>                                                                                                                                                                                                       |                                                                                                                                                                                                             |
| Calibration function                                                                                                                                                                                                                | shall pass the variability test defined in QAL2;<br>the range of testing should be 3 times the ELV                                                                                                          |
| Reproducibility under field test conditions                                                                                                                                                                                         | shall be less than 3 % of ELV                                                                                                                                                                               |
| Zero and span drifts                                                                                                                                                                                                                | shall be less than 3 % of ELV                                                                                                                                                                               |
| Availability and period of unattended operation                                                                                                                                                                                     | to be assessed during QAL1                                                                                                                                                                                  |

It shall be proven by use of EN ISO 14956 that the total uncertainty of the measured results obtained by the AMS meets the specifications for uncertainty stated in the applicable regulations. This total uncertainty is calculated by summing all the relevant uncertainty components arising from the individual performance characteristics in an appropriate manner. The AMS suitability shall be evaluated for the plant specific conditions taking into account the anticipated variations of the physical environment and interferences.

## 6 AMS location

The AMS shall be located in such a way that the gas sample which is monitored has a representative content of the particulate matter.

NOTE The location should be chosen taking into account the manufacturer recommendations. It may depend on the AMS principle of detection.

The AMS shall be located in such a way that there is minimum interference with the SRM and vice versa.

The working platform(s) for the AMS shall provide an easy and safe access to the AMS in order to allow frequent inspection and quality assurance procedure implementation.

The working platform for the SRM shall comply with the requirements of Annex A of EN 13284-1:2001 related to the manual gravimetric method.

## 7 Calibration and validation of the AMS (QAL2)

### 7.1 General

The AMS should be serviced just before the calibration according to the manufacturer's recommendations.

NOTE 1 It is recommended to include the service report as an appendix to the calibration report.

The AMS shall be calibrated and validated in accordance with EN 14181.

However, in the case where all measured results are below 30 % of the ELV and approval has been given by the competent authority, the number of measurements may be reduced to 5 measurements over 3 days. The total SRM sampling time shall be at least 7 h and 30 min (which is equal to 15 times 30 min) (see Annex B). If some of the measurements are above 30 % of the ELV, the calibration shall be extended to at least 15 valid measurements.

NOTE 2 Reducing the number of measurements and increasing the individual SRM sampling time leads to a better quantification of very low dust concentrations.

NOTE 3 Annex A shows an example of the application of QAL2.

### 7.2 Calibration procedure

#### 7.2.1 Parallel measurements with SRM

The standard reference method (SRM) used shall be in accordance with EN 13284-1. The testing laboratory, which performs the SRM measurements, shall have an accredited quality assurance system according to EN ISO/IEC 17025 or shall be approved directly by the competent authority.

As stated in EN 13284-1, if rinsing is required to recover the deposits upstream of the filter of the sampling equipment (see 8.5 and 10.5 of EN 13284-1), rinsing shall be performed after each measurement (SRM result and AMS reading) and not only after each measurement series.

## 7.2.2 Automatic recording of AMS output signals

### 7.2.2.1 General

The AMS output signal to be recorded may be analogue and/or digital. The AMS should be configured to have a minimum output signal of at least 5 times the detection limit and an output range which is large enough to gather readings with sufficient resolution.

The AMS output signal shall be automatically recorded during the parallel measurements in order to be electronically averaged. Data recording intervals and averaging times shall be reported specifically in the calibration report.

If it is proven that calibration curves do not change by switching from one range of the AMS to another, it may be calibrated in one range and used afterwards in another. This shall be documented in the calibration report.

Measures shall be taken to prevent AMS self-checks during SRM sampling.

### 7.2.2.2 Continuous AMS

The data sampling interval shall be less than half the AMS intrinsic averaging time.

### 7.2.2.3 Calibration of discontinuous AMS

Discontinuous AMS are characterized by a total measurement cycle of the AMS,  $t_{\text{cycle}}$ , and a sample time period,  $t_{\text{sample}}$ . The ratio  $t_{\text{sample}} / t_{\text{cycle}}$  should be larger than 90 %.

A ratio  $t_{\text{sample}} / t_{\text{cycle}}$  less than 90 % may be used, if there is no rapid change in the mass concentration of dust. These situations shall be approved by the competent authority.

The SRM measurement shall start at the beginning of a new AMS total measurement cycle and shall be performed for a complete number of measurement cycles.

## 7.3 Establishing the calibration function

The competent authority may approve the use of calibration procedure specified in 6.4.2 b) in EN 14181:2004, even if the range of the SRM results exceeds 15 % of ELV.

**NOTE** With dust AMS, it is advantageous to use the zero point in the calibration in order to avoid an inadequate calibration function.

## 7.4 Calibration under abnormal operating conditions

### 7.4.1 General

The dust concentration emissions are often close to zero compared with the ELV, if the dust cleaning equipment of the plant is operating normally. In such a case, the plant operator may decide to extend the calibration range by increasing the dust concentration by non operational means. The use of non operational means shall be agreed by both the plant operator and the competent authority and be completely documented in the calibration report.

**NOTE** For example, it may be necessary to slightly lower the efficiency of the gas cleaning equipment by simulating the more likely failures of the equipment, in order to increase the dust concentration up to the ELV.

Increasing the dust concentration by non operational means may lead to changes in dust characteristics and influence the calibration function of AMS which are cross-sensitive to particle size (optical devices, etc). In these cases, a quadratic calibration function may be used.

#### 7.4.2 Parallel measurements

The calibration range can be extended to the ELV by non operational means. In this case, at least 15 parallel measurements as described in EN13284-1 and EN 14181 for QAL2 are conducted in the extended calibration range such a way that:

- at least 5 measurements are performed in the normal operation range of the plant;
- at least 5 measurements are performed at the upper limit of the extended calibration range (by abnormal operating conditions);
- at least 5 measurements are performed at a dust concentration intermediate to these levels.

#### 7.4.3 Calibration function

If QAL1 shows that the AMS is sensitive to particle size, a quadratic calibration function may be calculated and used.

NOTE An example of a quadratic regression is shown in Annex C.

#### 7.4.4 Variability test

If a quadratic calibration function applies, the AMS measured values estimated with the quadratic calibration function shall be used in the variability test.

### 7.5 Special circumstances in fulfilling the variability test

#### 7.5.1 General

The AMS shall not be used to measure emissions according to this document, if the variability test is not passed. It is not allowed to selectively exclude single measurements, even if excess samples are taken, in order to fulfil the variability test.

#### 7.5.2 Exceptional conditions during calibration

If the operation of the plant changes to an exceptional condition during parallel measurements, it is allowed to exclude the corresponding data provided the required minimum number of parallel measurements is obtained. This shall be justified in the test report.

#### 7.5.3 Occurrence of semi volatile particulate matter

Emissions of some processes may contain semi-volatile compounds (i.e. in particulate form at low temperature and in gaseous form at higher temperature). In such a case, the measured dust concentration depends on the temperature. This may have a disturbing effect on the calibration, if the sampling temperature of the AMS (i.e. stack temperature for in situ AMS) differs from the SRM conventional temperature, usually 160 °C. If the variability test is not fulfilled due to the influence of semi-volatiles, other conventions may be adopted for the SRM temperature according to Clause 9 of EN 13284-1:2001. This shall be indicated in the calibration report.

If the gas stack temperature is at or below 110 °C, an efficient way of detecting and accounting for the semi-volatile particulate effect is:

- to set the SRM conventional temperature to 120 °C (i.e. filtration and drying temperature) and weigh the samples;
- to re-dry the collected dust at 160 °C, and re-weigh the samples;

— to compare the two calibration functions and the results of the corresponding variability tests.

If the variability test corresponding to the 160 °C drying temperature is not fulfilled, the calibration function from the lower drying temperature may be used, provided the variability test is passed. This shall be indicated in the calibration report.

If neither calibration function passes the variability test, the calibration as such does not pass the quality test and shall be rejected.

## **8 Ongoing quality assurance during operation (QAL3)**

QAL3 shall be performed according to EN 14181. For that, the AMS shall have appropriate zero and span reference materials which have been assessed as part of QAL1 (see Clause 5).

## **9 Annual surveillance test (AST)**

AST shall be performed according to EN 14181. If the calibration range is below 30 % of ELV (see 7.1), the number of parallel measurements may be reduced to 3.

## Annex A (informative)

### Example of calculation of the calibration function and of the variability test

#### A.1 General

The purpose of QAL2 is to perform a calibration, determine the variability and check for compliance with legal requirements on the measurement uncertainty.

In this example, AMS and SRM measurements have been performed at a municipal waste incinerator plant. The plant is equipped with an optical dust AMS (light scattering device), the electrical output of which has been registered during the measurements. SRM measurements have been performed according to EN 13284-1. Furthermore, the values of reference quantities have been measured with the plant and the SRM equipment as required by EN 14181.

The calibration function has been calculated and the variability been tested on the basis of these measurements according to EN 14181.

#### A.2 Data evaluation

##### A.2.1 General

Table A.1 lists the measurement methods and requirements used in this example.

**Table A.1 — Measurement methods and requirements**

| Parameter                                                                                                           | Particulate emission                       |
|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------|
| AMS method                                                                                                          | Optical dust AMS (light scattering device) |
| Offset for AMS                                                                                                      | 4 mA                                       |
| SRM method                                                                                                          | Manual, gravimetric – EN 13284-1           |
| Emission limit value (ELV) for standard conditions (0 °C, 1 013 hPa, dry gas, 11 % O <sub>2</sub> ) – Daily average | 10 mg/m <sup>3</sup>                       |
| Demands to percentage related to ELV                                                                                | 30 %                                       |

In this example, 15 parallel measurements have been taken over 3 days, evenly distributed. The results at AMS conditions are given in Table A.2:

Table A.2 — Measurements for the QAL2 test

| Sample number<br><i>i</i> | SRM value<br><i>y<sub>i</sub></i><br>mg/m <sup>3</sup> | AMS signal<br><i>x<sub>i</sub></i><br>mA |
|---------------------------|--------------------------------------------------------|------------------------------------------|
| 1                         | 4,05                                                   | 6,14                                     |
| 2                         | 8,69                                                   | 9,25                                     |
| 3                         | 2,49                                                   | 5,35                                     |
| 4                         | 4,62                                                   | 6,31                                     |
| 5                         | 5,68                                                   | 8,07                                     |
| 6                         | 2,17                                                   | 5,19                                     |
| 7                         | 2,80                                                   | 5,24                                     |
| 8                         | 2,56                                                   | 5,55                                     |
| 9                         | 3,28                                                   | 5,63                                     |
| 10                        | 3,69                                                   | 6,11                                     |
| 11                        | 2,55                                                   | 5,33                                     |
| 12                        | 3,95                                                   | 6,21                                     |
| 13                        | 1,21                                                   | 4,78                                     |
| 14                        | 1,25                                                   | 4,67                                     |
| 15                        | 0,85                                                   | 4,52                                     |

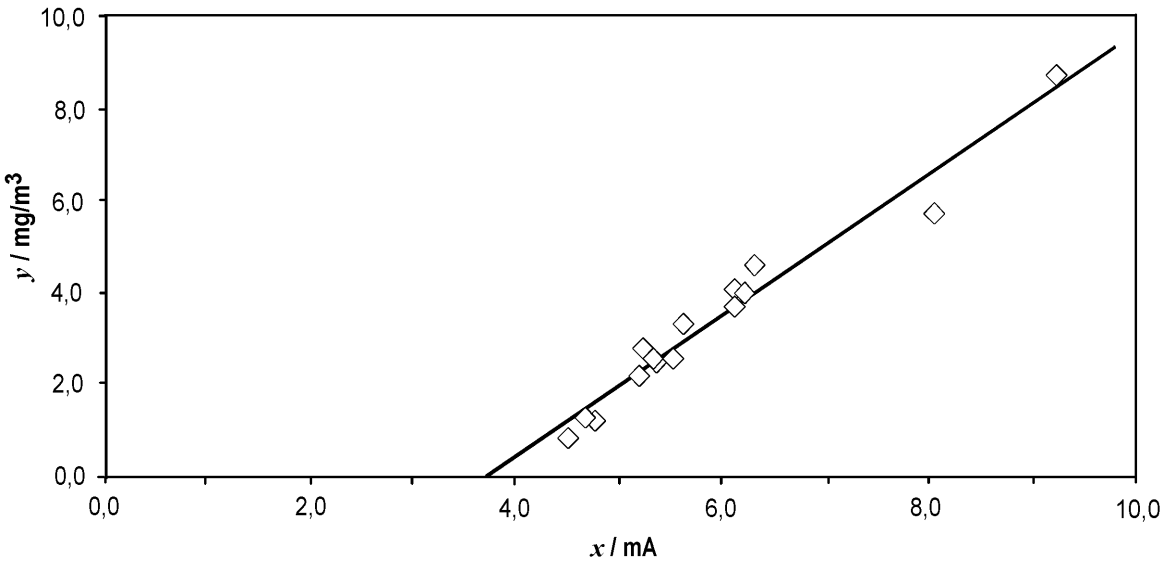


Figure A.1 — Plot of SRM measured values *y* versus AMS measured signals *x* and the calibration function determined in A.2.3



First the SRM results are plotted against the AMS values (see Figure A.1). A plot of the SRM results against the AMS signals is recommended to visualise the data and to help understand the results obtained by the statistical procedures.

### A.2.2 Calculation of measured values of the SRM at standard conditions

The requirement on the measurement uncertainty of the AMS measured values is expressed as 30 % of the daily ELV, which is given at standard conditions (0 °C, 1 013 hPa, dry flue gas with an oxygen content of 11 %).

To decide whether method a) or b) of 6.4.2 of EN 14161:2004 is used for calibrating the AMS, it is essential to calculate the SRM values at standard conditions. The values  $y$  of the SRM should therefore be converted to this standard condition in accordance with the equation (A.1) by using the appropriate parts:

$$y_s = y \times \frac{t + 273,15\text{K}}{273,15\text{K}} \times \frac{1013\text{hPa}}{1013\text{hPa} + p} \times \frac{100\%}{100\% - h} \times \frac{21\% - o_s}{21\% - o} \quad (\text{A.1})$$

where

- $t$  is the Celsius temperature;
- $p$  is the difference between the static pressure of the sample gas and the standard pressure;
- $h$  is the absolute water vapour content (by volume);
- $o$  is the oxygen content in dry gas (by volume);
- $o_s$  is the oxygen standard condition.

The peripheral measurements used to perform the conversion of the SRM values and the results are given in Table A.3.

Table A.3 — QAL2 measurements by the SRM

| Sample number<br><i>i</i> | SRM measured value<br><i>y<sub>i</sub></i><br>mg/m <sup>3</sup> | Temperature<br><i>t<sub>i</sub></i><br>°C | Moisture content<br><i>h<sub>i</sub></i><br>% | O <sub>2</sub> -content in dry flue gas<br><i>o<sub>i</sub></i><br>% | SRM value at standard conditions<br><i>y<sub>i,s</sub></i><br>mg/m <sup>3</sup> |
|---------------------------|-----------------------------------------------------------------|-------------------------------------------|-----------------------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 1                         | 4,05                                                            | 143,0                                     | 15,1                                          | 12,4                                                                 | 8,46                                                                            |
| 2                         | 8,69                                                            | 144,0                                     | 15,6                                          | 12,9                                                                 | 19,42                                                                           |
| 3                         | 2,49                                                            | 145,0                                     | 14,3                                          | 12,3                                                                 | 5,11                                                                            |
| 4                         | 4,62                                                            | 146,0                                     | 14,9                                          | 12,2                                                                 | 9,47                                                                            |
| 5                         | 5,68                                                            | 145,0                                     | 14,7                                          | 12,4                                                                 | 11,85                                                                           |
| 6                         | 2,17                                                            | 144,0                                     | 16,2                                          | 11,1                                                                 | 4,00                                                                            |
| 7                         | 2,80                                                            | 144,0                                     | 16,3                                          | 11,7                                                                 | 5,49                                                                            |
| 8                         | 2,56                                                            | 142,0                                     | 16,0                                          | 12,1                                                                 | 5,20                                                                            |
| 9                         | 3,28                                                            | 143,0                                     | 16,1                                          | 11,6                                                                 | 6,34                                                                            |
| 10                        | 3,69                                                            | 144,0                                     | 14,4                                          | 12,3                                                                 | 7,57                                                                            |
| 11                        | 2,55                                                            | 144,0                                     | 16,1                                          | 11,7                                                                 | 4,99                                                                            |
| 12                        | 3,95                                                            | 145,0                                     | 15,2                                          | 12,1                                                                 | 8,01                                                                            |
| 13                        | 1,21                                                            | 143,0                                     | 16,9                                          | 12,0                                                                 | 2,47                                                                            |
| 14                        | 1,25                                                            | 145,0                                     | 14,9                                          | 12,4                                                                 | 2,61                                                                            |
| 15                        | 0,85                                                            | 146,0                                     | 15,7                                          | 11,9                                                                 | 1,70                                                                            |

### A.2.3 Calibration function

The calibration function is determined at AMS operating conditions. It is described by equation (A.2):

$$y_i = a + b x_i \quad (\text{A.2})$$

where

$x_i$  is the result from the AMS;

$y_i$  is the result from the SRM;

$a$  is the intercept;

$b$  is the slope.

From Table A.3 it can be seen for the maximum and minimum value at standard conditions that:

$$y_{s,\max} = 19,4 \text{ mg/m}^3$$

$$y_{s,\min} = 1,7 \text{ mg/m}^3$$

$$y_{s,\max} - y_{s,\min} = 17,7 \text{ mg/m}^3$$

To investigate if the difference ( $y_{s,\max} - y_{s,\min}$ ) is smaller than 15 % of the ELV, the following calculation is carried out, where the ELV is based on standard conditions:

$$\begin{aligned}
 \Delta y_{\max} &= 0,15 \text{ E} \\
 &= 0,15 \times 10 \text{ mg/m}^3 \\
 &= 1,5 \text{ mg/m}^3
 \end{aligned}$$

The difference is:

$$y_{s,\max} - y_{s,\min} = 17,7 \text{ mg/m}^3 > \Delta y_{\max} = 0,15 \text{ E} = 1,5 \text{ mg/m}^3$$

Since the difference is greater than 15 % of ELV,  $a$  and  $b$  are calculated by:

$$\hat{b} = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2}$$

$$\hat{a} = \bar{y} - \hat{b} \bar{x}$$

where

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i = \frac{1}{15} \times 88,36 \text{ mA} = 5,89 \text{ mA}$$

$$\bar{y} = \frac{1}{N} \sum_{i=1}^N y_i = \frac{1}{15} \times 49,85 \frac{\text{mg}}{\text{m}^3} = 3,32 \frac{\text{mg}}{\text{m}^3}$$

The slope and intercept are calculated by:

$$\hat{b} = 1,53 \frac{\text{mg/m}^3}{\text{mA}}$$

$$\hat{a} = -5,68 \frac{\text{mg}}{\text{m}^3}$$

The calibration function then becomes (see Figure A.1):

$$\hat{y}_i = \hat{a} + \hat{b} x_i = -5,68 \frac{\text{mg}}{\text{m}^3} + 1,53 \frac{\text{mg/m}^3}{\text{mA}} x_i \quad (\text{A.3})$$

#### A.2.4 Calibrated values of AMS

The results listed in Table A.4 are obtained by using the calibration function given by equation (A.3) on the AMS measured signals:

Table A.4 — Calibration of AMS

| Number<br>$i$ | AMS<br>measured signal<br>$x_i$<br>mA | SRM<br>measured value<br>$y_i$<br>mg/m <sup>3</sup> | AMS<br>measured value<br>$\hat{y}_i$<br>mg/m <sup>3</sup> |
|---------------|---------------------------------------|-----------------------------------------------------|-----------------------------------------------------------|
| 1             | 6,14                                  | 4,05                                                | 3,70                                                      |
| 2             | 9,25                                  | 8,69                                                | 8,47                                                      |
| 3             | 5,35                                  | 2,49                                                | 2,50                                                      |
| 4             | 6,31                                  | 4,62                                                | 3,97                                                      |
| 5             | 8,07                                  | 5,68                                                | 6,65                                                      |
| 6             | 5,19                                  | 2,17                                                | 2,25                                                      |
| 7             | 5,24                                  | 2,80                                                | 2,34                                                      |
| 8             | 5,55                                  | 2,56                                                | 2,80                                                      |
| 9             | 5,63                                  | 3,28                                                | 2,92                                                      |
| 10            | 6,11                                  | 3,69                                                | 3,66                                                      |
| 11            | 5,33                                  | 2,55                                                | 2,47                                                      |
| 12            | 6,21                                  | 3,95                                                | 3,82                                                      |
| 13            | 4,78                                  | 1,21                                                | 1,63                                                      |
| 14            | 4,67                                  | 1,25                                                | 1,45                                                      |
| 15            | 4,52                                  | 0,85                                                | 1,23                                                      |

### A.2.5 Conversion of data to standard conditions

The requirements on the quality of the AMS is expressed as 30 % of the daily ELV, which is given at standard conditions (0 °C, 1 013 hPa, dry flue gas with an oxygen content of 11 %). The values of the SRM and the calibrated AMS should therefore be converted to these standard conditions. Equation (A.1) is used for the conversion of SRM values. The conversion of AMS values is carried out by application of the corresponding equation (A.4) by:

$$\hat{y}_s = \hat{y} \times \frac{t + 273,15\text{K}}{273,15\text{K}} \times \frac{1013\text{hPa}}{1013\text{hPa} + p} \times \frac{100\%}{100\% - h} \times \frac{21\% - o_s}{21\% - o} \quad (\text{A.4})$$

where

- $t$  is the Celsius temperature;
- $p$  is the difference between the static pressure of the sample gas and the standard pressure;
- $h$  is the absolute water vapour content (by volume);
- $o$  is the oxygen content in dry gas (by volume);
- $o_s$  is the oxygen standard condition.

The peripheral measurements used to perform the conversion of the AMS-values and the results are given in Table A.5:

Table A.5 — QAL2 measurements by AMS

| Sample number<br><br>$i$ | AMS measured signal<br><br>$x_i$<br>mA | AMS calibrated value<br><br>$\hat{y}_i$<br>mg/m <sup>3</sup> | Temperature<br><br>$t_i$<br>°C | Moisture content (calculated)<br><br>$h_i$<br>% | O <sub>2</sub> -content in dry flue gas<br><br>$o_i$<br>% | AMS calibrated value at standard conditions<br><br>$\hat{y}_{i,s}$<br>mg/m <sup>3</sup> |
|--------------------------|----------------------------------------|--------------------------------------------------------------|--------------------------------|-------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------|
| 1                        | 6,14                                   | 3,70                                                         | 149,0                          | 15,0                                            | 12,0                                                      | 7,47                                                                                    |
| 2                        | 9,25                                   | 8,47                                                         | 143,0                          | 15,0                                            | 12,0                                                      | 16,86                                                                                   |
| 3                        | 5,35                                   | 2,50                                                         | 146,0                          | 15,0                                            | 12,0                                                      | 5,02                                                                                    |
| 4                        | 6,31                                   | 3,97                                                         | 146,0                          | 15,0                                            | 12,0                                                      | 7,97                                                                                    |
| 5                        | 8,07                                   | 6,65                                                         | 146,0                          | 15,0                                            | 12,0                                                      | 13,34                                                                                   |
| 6                        | 5,19                                   | 2,25                                                         | 147,0                          | 15,0                                            | 11,6                                                      | 4,33                                                                                    |
| 7                        | 5,24                                   | 2,34                                                         | 147,0                          | 15,0                                            | 11,9                                                      | 4,64                                                                                    |
| 8                        | 5,55                                   | 2,80                                                         | 148,0                          | 15,0                                            | 12,2                                                      | 5,76                                                                                    |
| 9                        | 5,63                                   | 2,92                                                         | 148,0                          | 15,0                                            | 12,0                                                      | 5,89                                                                                    |
| 10                       | 6,11                                   | 3,66                                                         | 147,0                          | 15,0                                            | 12,5                                                      | 7,80                                                                                    |
| 11                       | 5,33                                   | 2,47                                                         | 147,0                          | 15,0                                            | 12,1                                                      | 5,02                                                                                    |
| 12                       | 6,21                                   | 3,82                                                         | 147,0                          | 15,0                                            | 12,3                                                      | 7,94                                                                                    |
| 13                       | 4,78                                   | 1,63                                                         | 146,0                          | 15,0                                            | 12,3                                                      | 3,37                                                                                    |
| 14                       | 4,67                                   | 1,45                                                         | 146,0                          | 15,0                                            | 12,0                                                      | 2,91                                                                                    |
| 15                       | 4,52                                   | 1,23                                                         | 145,0                          | 15,0                                            | 12,0                                                      | 2,47                                                                                    |

### A.2.6 Valid calibration range

The calibration function is valid in the range from zero to  $\hat{y}_{s,\max}$  plus an extension of 10 %. For this AMS, the value is  $\hat{y}_{s,\max} = 16,86 \text{ mg/m}^3$  and therefore the valid calibration range is:

$$0 \leq \hat{y}_s \leq 1,10 \times 16,86 \text{ mg/m}^3$$

$$0 \leq \hat{y}_s \leq 18,55 \text{ mg/m}^3$$

The valid calibration range is from 0,0 mg/m<sup>3</sup> to 18,6 mg/m<sup>3</sup> at standard conditions.

### A.2.7 Test of variability

The variability test is performed on standard conditions and is accepted if:

$$s_D \leq \sigma_0 k_v \quad (\text{A.5})$$

where

$s_D$  is the standard deviation of the differences  $D_i$ ;

$\sigma_0$  is the uncertainty laid down by the authorities;

$k_v$  is the test parameter.

The standard deviation  $s_D$  is calculated in accordance with equation (A.6):

$$s_D = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (D_i - \bar{D})^2} \quad (\text{A.6})$$

where the difference  $D_i$  is:

$$D_i = y_{i,s} - \hat{y}_{i,s} \quad (\text{A.7})$$

The values used for the calculation of  $s_D$  are given in Table A.6:

**Table A.6 — Figures used for variability test at standard conditions <sup>a</sup>**

| Sample number<br><br>$i$                                                        | AMS calibrated value at standard conditions <sup>A</sup><br><br>$\hat{y}_{i,s}$<br><br>mg/m <sup>3</sup> | SRM value at standard conditions <sup>a</sup><br><br>$y_{i,s}$<br><br>mg/m <sup>3</sup> | Difference<br><br>$D_i = y_{i,s} - \hat{y}_{i,s}$<br><br>mg/m <sup>3</sup> | Difference<br><br>$D_i - \bar{D}$<br><br>mg/m <sup>3</sup> | Squared difference<br><br>$(D_i - \bar{D})^2$<br><br>(mg/m <sup>3</sup> ) <sup>2</sup> |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------|------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 1                                                                               | 7,47                                                                                                     | 8,46                                                                                    | 0,989                                                                      | 0,863                                                      | 0,744                                                                                  |
| 2                                                                               | 16,86                                                                                                    | 19,42                                                                                   | 2,560                                                                      | 2,434                                                      | 5,922                                                                                  |
| 3                                                                               | 5,02                                                                                                     | 5,11                                                                                    | 0,089                                                                      | −0,038                                                     | 0,001                                                                                  |
| 4                                                                               | 7,97                                                                                                     | 9,47                                                                                    | 1,505                                                                      | 1,378                                                      | 1,899                                                                                  |
| 5                                                                               | 13,34                                                                                                    | 11,85                                                                                   | −1,489                                                                     | −1,615                                                     | 2,609                                                                                  |
| 6                                                                               | 4,33                                                                                                     | 4,00                                                                                    | −0,329                                                                     | −0,456                                                     | 0,208                                                                                  |
| 7                                                                               | 4,64                                                                                                     | 5,49                                                                                    | 0,845                                                                      | 0,719                                                      | 0,516                                                                                  |
| 8                                                                               | 5,76                                                                                                     | 5,20                                                                                    | −0,564                                                                     | −0,690                                                     | 0,477                                                                                  |
| 9                                                                               | 5,89                                                                                                     | 6,34                                                                                    | 0,448                                                                      | 0,321                                                      | 0,103                                                                                  |
| 10                                                                              | 7,80                                                                                                     | 7,57                                                                                    | −0,228                                                                     | −0,355                                                     | 0,126                                                                                  |
| 11                                                                              | 5,02                                                                                                     | 4,99                                                                                    | −0,025                                                                     | −0,152                                                     | 0,023                                                                                  |
| 12                                                                              | 7,94                                                                                                     | 8,01                                                                                    | 0,071                                                                      | −0,056                                                     | 0,003                                                                                  |
| 13                                                                              | 3,37                                                                                                     | 2,47                                                                                    | −0,904                                                                     | −1,031                                                     | 1,063                                                                                  |
| 14                                                                              | 2,91                                                                                                     | 2,61                                                                                    | −0,299                                                                     | −0,426                                                     | 0,181                                                                                  |
| 15                                                                              | 2,47                                                                                                     | 1,70                                                                                    | −0,768                                                                     | −0,895                                                     | 0,801                                                                                  |
| Average                                                                         |                                                                                                          |                                                                                         | 0,127                                                                      |                                                            |                                                                                        |
| Sum                                                                             |                                                                                                          |                                                                                         |                                                                            |                                                            | 14,676                                                                                 |
| <sup>a</sup> Standard conditions: 0 °C, 1 013 hPa, dry gas, 11 % O <sub>2</sub> |                                                                                                          |                                                                                         |                                                                            |                                                            |                                                                                        |

The standard deviation  $s_D$  is given by:

$$s_D = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (D_i - \bar{D})^2} = \sqrt{\frac{1}{15-1} 14,68 \left( \frac{\text{mg}}{\text{m}^3} \right)^2} = 1,02 \frac{\text{mg}}{\text{m}^3}$$

The uncertainty laid down by the authorities is 30 % of the ELV as a 95 % confidence interval.  $\sigma_0$  is therefore calculated as:

$$\begin{aligned} \sigma_0 &= p E / 1,96 \\ &= \frac{0,30 \times 10 \text{ mg/m}^3}{1,96} \\ &= 1,53 \text{ mg/m}^3 \end{aligned}$$

For 15 measurements the  $k_v$  value is 0,9761 (see Table 1 in EN 14181:2004). The test for variability then results in:

$$\begin{aligned} 1,02 \text{ mg/m}^3 &< 1,53 \text{ mg/m}^3 \times 0,9761 \\ 1,02 \text{ mg/m}^3 &< 1,49 \text{ mg/m}^3 \end{aligned}$$

which is fulfilled. The AMS passes the test.

## Annex B (informative)

### Number of measurement points for calibration

#### B.1 General

In cases where the calibration range is less than the uncertainty specified in the relevant EU Directives (e.g. 30 % of ELV) it is allowed according to 7.1 to use fewer measurement points for the following reasons:

- fewer SRM points taken for a proportionally longer duration usually result in less total error;
- the total uncertainty cannot exceed the amount allowed by the EU Directive due to the limited range;
- the calibration range does not include the emission limit.

#### B.2 Statistical effect of using fewer measurement points

The test values  $k_v$  for the variability test according to EN 14181 (based on a  $\chi^2$ -test with a  $\beta$ -value of 50 % and for  $N$  numbers of paired measurements) are given in Table B.1.

**Table B.1 —  $k_v$  values**

| Number of measurements $N$ | $k_v$   |
|----------------------------|---------|
| 15                         | 0,976 1 |
| 8                          | 0,952 1 |
| 5                          | 0,916 1 |

#### B.3 Errors in calibration at low dust concentration

On making calibrations at low dust concentrations (the focus of EN 13284), one obtains less total uncertainty by focussing on more accurate (and longer) sampling to reduce the standard deviation by minimising uncertainties in the SRM, rather than necessarily testing larger number of samples. For example, uncertainty caused by short period tests (collecting insufficient weight of dust), might increase errors by a factor of 10 or 20 (see 12.2 and 12.3 of EN 13284-1:2001) whereas using 15 samples vs 5 samples has a relatively small effect on the variability (see above).



## Annex C (informative)

### Quadratic regression

The calculation of the coefficients  $A$ ,  $B$  and  $C$  of the quadratic regression curve given in equation (C.1) is carried out according to the Gaussian multiplication:

$$\hat{y} = A x^2 + B x + C \quad (\text{C.1})$$

The matrix  $(C_{ij})$  of the multiplier is given by equation (C.2):

$$(C_{ij}) = \begin{pmatrix} C_{00} & C_{01} & C_{02} \\ C_{10} & C_{11} & C_{12} \\ C_{20} & C_{21} & C_{22} \end{pmatrix} \quad (\text{C.2})$$

The values  $C_{ij}$  are derived from the following set of equations (C.3), (C.4) and (C.5):

$$\begin{aligned} s_{00} C_{00} + s_{01} C_{10} + s_{02} C_{20} &= 1 \\ s_{10} C_{00} + s_{11} C_{10} + s_{12} C_{20} &= 0 \\ s_{20} C_{00} + s_{21} C_{10} + s_{22} C_{20} &= 0 \end{aligned} \quad (\text{C.3})$$

$$\begin{aligned} s_{00} C_{01} + s_{01} C_{11} + s_{02} C_{21} &= 0 \\ s_{10} C_{01} + s_{11} C_{11} + s_{12} C_{21} &= 1 \\ s_{20} C_{01} + s_{21} C_{11} + s_{22} C_{21} &= 0 \end{aligned} \quad (\text{C.4})$$

$$\begin{aligned} s_{00} C_{02} + s_{01} C_{12} + s_{02} C_{22} &= 0 \\ s_{10} C_{02} + s_{11} C_{12} + s_{12} C_{22} &= 0 \\ s_{20} C_{02} + s_{21} C_{12} + s_{22} C_{22} &= 1 \end{aligned} \quad (\text{C.5})$$

where

$$\begin{aligned} s_{00} &= N \\ s_{01} = s_{10} &= \sum_{k=1}^N x_k \\ s_{02} = s_{20} = s_{11} &= \sum_{k=1}^N x_k^2 \\ s_{21} = s_{12} &= \sum_{k=1}^N x_k^3 \\ s_{22} &= \sum_{k=1}^N x_k^4 \end{aligned} \quad (\text{C.6})$$

In addition, the following equations (C.7) are needed later (see equation (C.10)):

$$\begin{aligned}
s_0 &= \sum_{k=1}^N y_k \\
s_1 &= \sum_{k=1}^N x_k y_k \\
s_2 &= \sum_{k=1}^N x_k^2 y_k
\end{aligned} \tag{C.7}$$

The determinant  $D$  for the calculation of the Gaussian multipliers is given by equation (C.8):

$$\begin{aligned}
D &= \begin{vmatrix} s_{00} & s_{01} & s_{02} \\ s_{10} & s_{11} & s_{12} \\ s_{20} & s_{21} & s_{22} \end{vmatrix} \\
&= \begin{vmatrix} s_{00} & s_{01} & s_{02} \\ s_{01} & s_{02} & s_{21} \\ s_{02} & s_{21} & s_{22} \end{vmatrix} \\
&= s_{00} \cdot (s_{02} s_{22} - s_{21}^2) + s_{01} (s_{21} s_{02} - s_{01} s_{22}) + s_{02} (s_{01} s_{21} - s_{02}^2)
\end{aligned} \tag{C.8}$$

The Gaussian multipliers given in equation (C.9) are obtained using Kramer's rule:

$$\begin{aligned}
C_{00} &= \frac{s_{02} s_{22} - s_{21}^2}{D} \\
C_{10} &= \frac{s_{21} s_{02} - s_{01} s_{22}}{D} = C_{01} \\
C_{20} &= \frac{s_{01} s_{21} - s_{02}^2}{D} = C_{02} \\
C_{11} &= \frac{s_{00} s_{22} - s_{02}^2}{D} \\
C_{21} &= \frac{s_{01} s_{02} - s_{00} s_{21}}{D} = C_{12} \\
C_{22} &= \frac{s_{00} s_{02} - s_{01}^2}{D}
\end{aligned} \tag{C.9}$$

The values  $A$ ,  $B$  and  $C$  are obtained by equation (C.10):

$$\begin{aligned}
A &= C_{20} s_0 + C_{21} s_1 + C_{22} s_2 \\
B &= C_{10} s_0 + C_{11} s_1 + C_{12} s_2 \\
C &= C_{00} s_0 + C_{01} s_1 + C_{02} s_2
\end{aligned} \tag{C.10}$$

## Bibliography

- [1] Directive 2000/76/EC of the European Parliament and of the Council of 4 December 2000 on the incineration of waste
- [2] Directive 2001/80/EC of the European Parliament and of the Council of 23 October 2001 on the limitation of emissions of certain pollutants into the air from large combustion plants

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